# Metadome AI — Home Decor Assistant

**README** — Product Manager Role

## Overview

A lightweight Unity prototype that demonstrates a minimal product vertical for AI-assisted interior suggestions. The prototype lets a user type a style (e.g., **modern**, **minimalist**), generates 3 curated options, previews them by spawning real 3D prefabs (sofa, chair, coffee table) and applies material variations. Users can then drag furniture around the room to try layouts.

This document summarizes what was implemented, how to run the project, what files matter, screenshots that show the flow, and suggested next steps for product improvements.

## One-line value proposition

Help users quickly visualise furniture and material options by turning a short style prompt into 3 concrete design previews that they can place in a room.

## Key features

- **Simulated generative AI suggestions** — type a style keyword and the app shows 3 curated option buttons (example: *Light Modern*, *Dark Modern*, *Luxury Modern*).
- **Spawn actual prefabs** — picking an option spawns 3 prefabs (sofa, chair, coffee table) in the scene so users see real 3D objects, not only names.
- **Material presets** — selecting an option applies a preset material to the spawned models so the user sees a true preview.
- **Drag & drop placement** — spawned objects include a `FurnitureDrag` script enabling click-and-drag placement on the floor.
- **Manual controls** — quick buttons at the top allow users to add single objects manually (Chair, Sofa, CoffeeTable).
- **Inspector-first wiring** — all references (prefabs, materials, UI buttons) are exposed in `FurnitureManager` and `AISuggester` for easy tuning by designers.

## How to run :

- Open the project in the Unity Editor and open the **MainScene**.
- Select the `GameManager` GameObject in the Hierarchy and inspect the `FurnitureManager` and `AISuggester` components.
- Assign **Sofa Prefab**, **Table Prefab**, **Chair Prefab**.
- Assign the material presets for each style group (Modern / Minimal / Classic) inside `FurnitureManager`.
- In `AISuggester`, assign **Style Input** (TMP Input), **Suggestion Text**, and **Option1/Option2/Option3** (UI Buttons).

**Press Play:**

- Type a style (e.g., `modern`) and press **Generate Suggestions**.
- Click one of the three option buttons to spawn the set and apply the selected material.
- Use the top buttons (Chair / Sofa / CoffeeTable) to add single objects.
- Click and drag spawned objects to reposition them on the floor

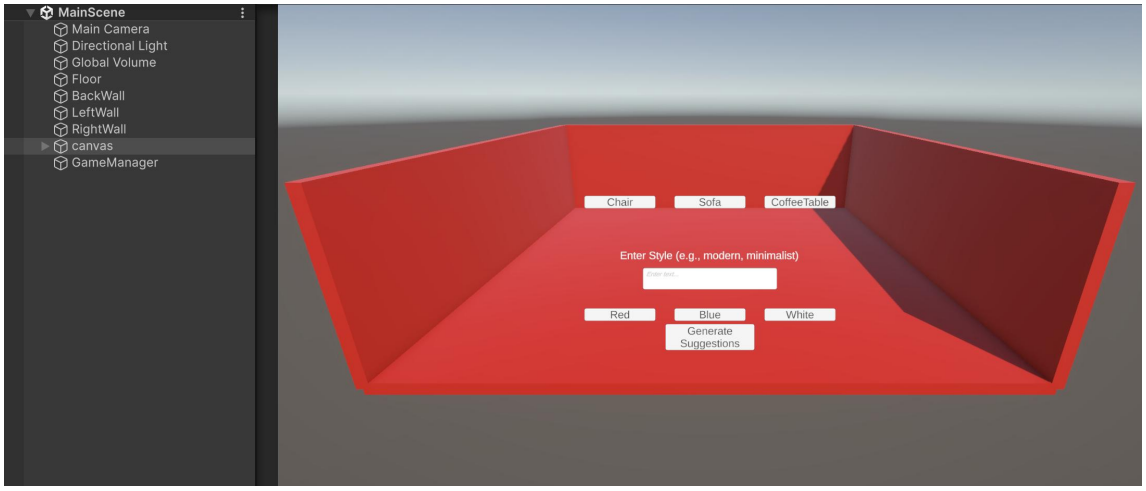
# Important files / scripts

- `AISuggester.cs` — maps style input to three option buttons and wires them to `FurnitureManager` methods.
- `FurnitureManager.cs` — spawns prefabs, clears previous set, and applies materials. Also adds `FurnitureDrag` to spawned objects
- `FurnitureDrag.cs` — simple InputSystem-based raycast drag behaviour for moving objects on the floor.

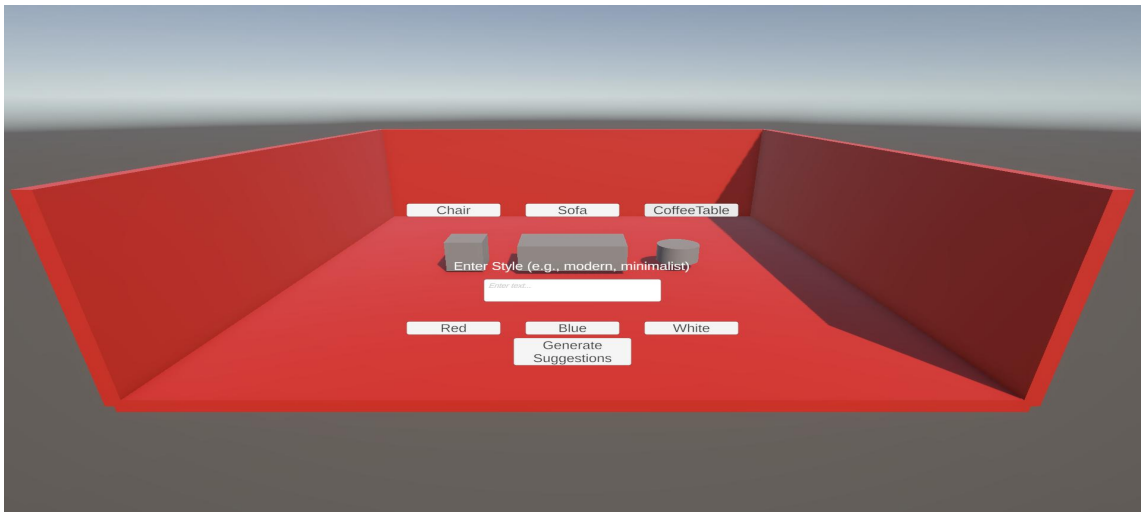
These three files are the core of the prototype and are included in the upload.

# Screenshots

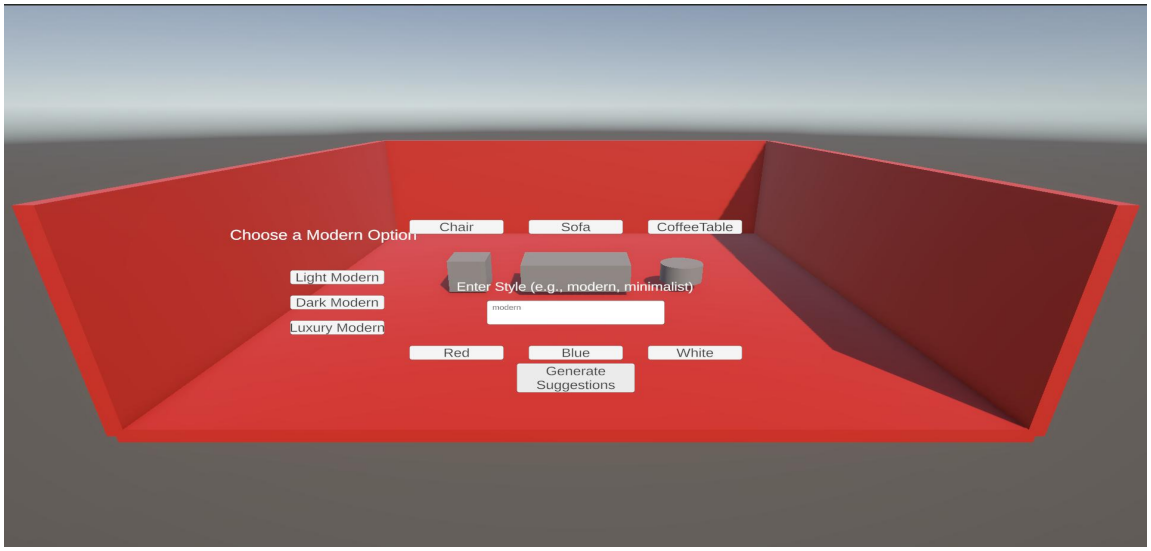
### Cover / Scene (initial)



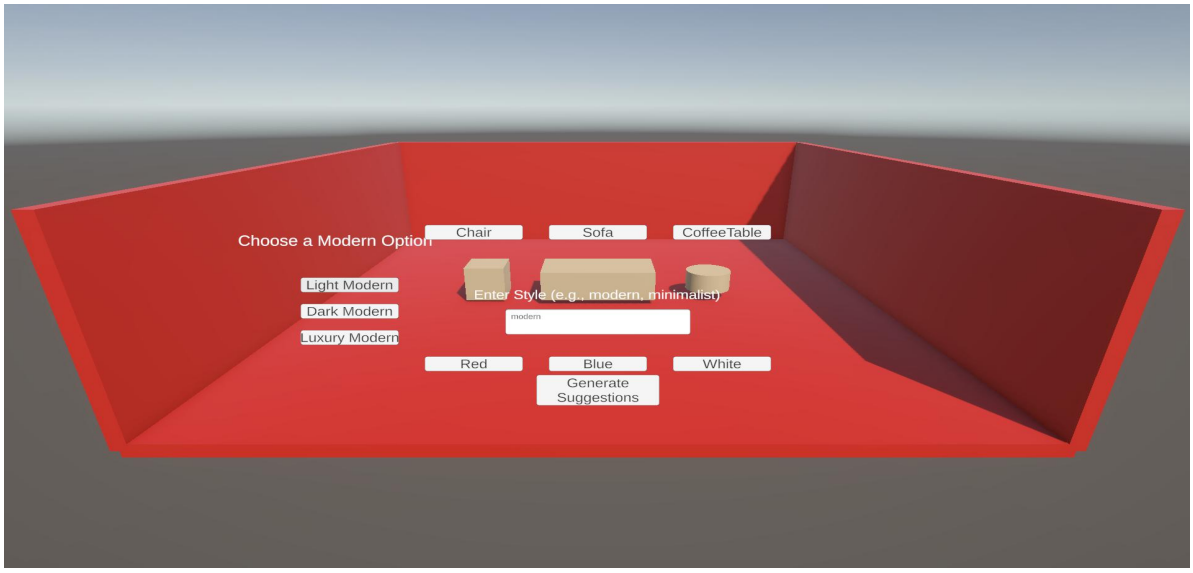

### Spawned furniture set (after pressing an option)



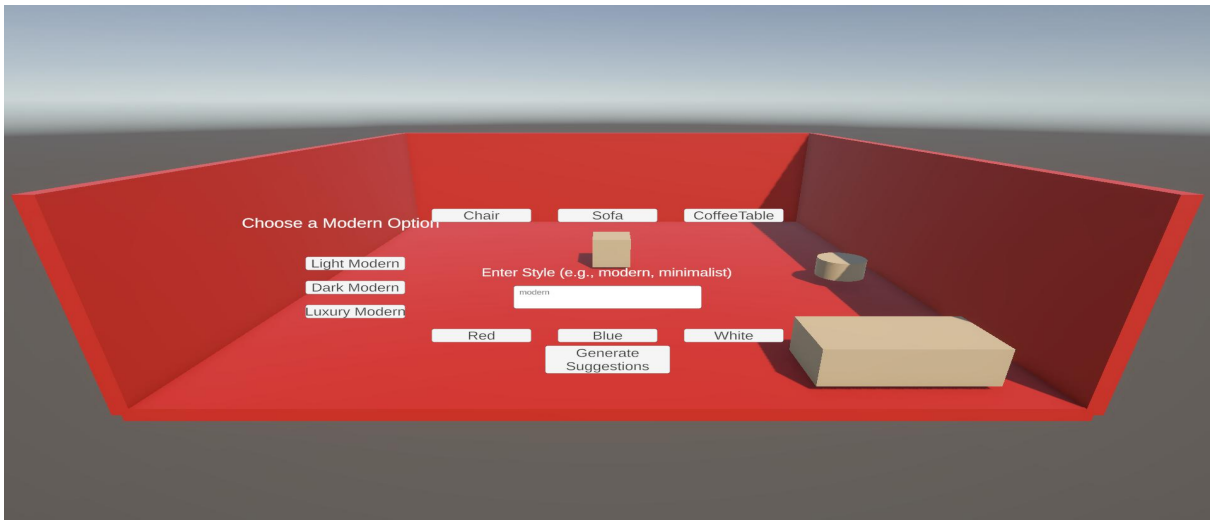
## Generated options (left-side option buttons)created by AISuggester)



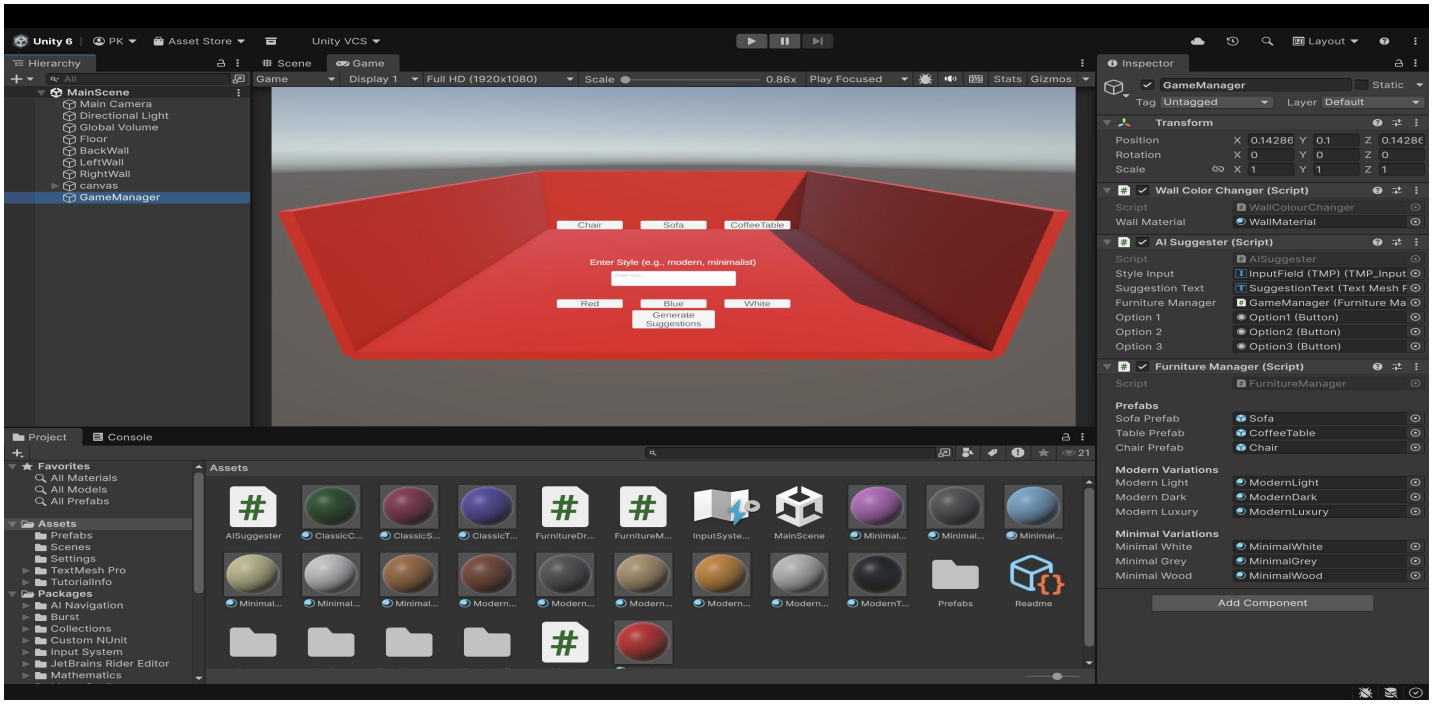

## Applied material preview




## Drag example



**Inspector: script links and assignments**




# What I completed

- Implemented core mechanics: spawn prefabs, clear room, apply materials, and drag placement.
- Built a small AI-suggester module that generates 3 curated options from textual style input.
- Exposed configuration points in inspector for a designer (prefabs, materials, UI links).
- Added manual spawn buttons and ensured prefabs have the drag component.

# Where to find things (project structure)

- `Assets/Prefabs/` — Sofa, Chair, CoffeeTable
- `Assets/Scripts/` — `AISuggester.cs`, `FurnitureManager.cs`, `FurnitureDrag.cs`
- `Assets/Materials/` — grouped materials (Modern, Minimal)
- `MainScene` — demo scene